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SYSTEMS AND METHODS FOR ACTIVE ASSAILANT RESPONSE

Abstract

An active assailant response system comprising a set of response units configured to deploy a nonlethal deterrent and including a first response unit arranged in a first zone of an environment and a second response unit arranged in a second zone of the environment; a set of detection units configured to identify one or more active assailants within the environment and including a first detection unit arranged in the first zone of the environment and a second detection unit arranged in the second zone of the environment; and a computer in communication with the set of response units and being configured to determine that at least one active assailant is present within one of the zones based on signals received from the set of detection units, and activate the set of response units in response to the at least one active assailant being present within one zones.

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Background/Summary

PRIORITY CLAIM [0001] This application claims priority to and all the benefits of U.S. Provisional Patent Application No. 63/563,574, filed on Mar. 11, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD
[0002] The present disclosure generally relates to systems and methods for responding to an active assailant.
BACKGROUND
[0003] Known active assailant responding devices and systems rely on human piloting of a response unit. This requires a launch sequence to enable the human pilot to be in control of the device on scene. Conventional launch sequences use valuable time that could instead be used to save lives. Other systems exist to put either nonlethals or a lethal weapon in the hand of an individual through a series of pre-installed boxes in an environment, such as a building. However, none of these systems can autonomously respond to the active assailant or operate within a mapped environment. As such, there is a need in the art for an active assailant response system that provides quick and autonomous response to active assailant situations, such as in an environment where the system has been pre-installed ahead of time.
SUMMARY
[0004] An active assailant response system is provided. The active assailant response system includes a set of response units, a set of detection units, and a computer in communication with the set of response units and the set of detection units. The set of response units is configured to deploy a nonlethal deterrent and includes a first response unit arranged in a first zone of an environment and a second response unit arranged in a second zone of the environment. The set of detection units is configured to identify one or more active assailants within the environment and includes a first detection unit arranged in the first zone of the environment and a second detection unit arranged in the second zone of the environment. The computer is configured to determine that at least one active assailant is present within one of the first and second zones based on signals received from the set of detection units and activate the set of response units in response to the at least one active assailant being present within one of the first and second zones.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.
[0006] FIG. **1** includes a perspective view of an active assailant response system according to one implementation.
[0007] FIG. **2** illustrates an arrangement of response unit relative to various zones of a school according to one implementation.
[0008] FIG. **3** includes a block diagram illustrating various components of the active assailant response system of FIG. **1**.

[0009] FIGS. **4**A-**4**C depict various perspective views of a turret unit according to one implementation.
[0010] FIGS. **5**A-**5**F depict various perspective views of a turret unit according to another implementation.
[0011] FIGS. **6**A-**6**D depict various perspective views of a rail traversal unit according to one implementation.
[0012] FIGS. **7**A-**7**D depict various perspective views of an aerial traversal unit according to one implementation.
[0013] FIGS. **8**A-**8**D depict various perspective views of a cable traversal unit according to one implementation.
[0014] FIGS. **9**A-**9**D depict various perspective views of a wheeled traversal unit according to one implementation.
[0015] FIG. **10** includes a block diagram showing connections between components of a response unit according to one implementation.
[0016] FIG. **11** includes a perspective view of a physical switch according to one implementation.
[0017] FIG. **12** depicts a graphical user interface of the activate assailant response system of FIG. **1** according to one implementation.
[0018] FIG. **13** includes a flowchart describing a method of controlling an active assailant response system according to a first implementation.
[0019] FIG. **14** includes a flowchart describing a method of controlling an active assailant response system according to a second implementation.
[0020] FIG. **15** includes a flowchart describing a method of controlling a response unit according to one implementation.
[0021] FIG. **16** includes a flowchart describing a method of updating response units according to one implementation.
[0022] FIG. **17** shows a first arrangement of response unit disposed in various zones of a small school.
[0023] FIG. **18** shows a second arrangement of response unit disposed in various zones of a large school.

DETAILED DESCRIPTION

[0024] Referring to the Figures, wherein like numerals indicate like or corresponding parts throughout the several views, an active assailant response system is shown throughout.

I. System Overview

[0025] The embodiments of the invention, as disclosed herein, offer optimal architecture and operation of an autonomous active assailant response system. The preferred implementations are illustrated in the context of autonomous active shooter response. A person of ordinary skill in the art will readily appreciate that the systems and methods disclosed herein may also be applicable to a number of other contexts wherein an active shooter, mass shooter, gunman, or other type of active assailant may exist as a threat to other individuals.
[0026] Referring to FIG. **1**, a perspective view of an active assailant response system **100** according to one implementation is shown. The system **100** may include at least one response unit **102** configured to deploy a nonlethal deterrent in response to an active assailant being detected/identified within an environment. The environment may include a building, a courtyard, an outdoor location, or a combination thereof. In the figures, the environment is shown as schools of various sizes, but the system **100** may be disposed within the other aforementioned types of environments. As shown, the response unit(s) **102** may traverse the environment via rail, cable, air, and/or ground. In some implementations, response units **102** of multiple types may be provided to increase the efficacy of the system **100**, like by reducing the chance that limitations of one type of traversal limits the ability of the system **100** to neutralize an active assailant. In order to control the unit(s) **102**, the system **100** may further include a response computer **104**. Further, the system **100**

may include an interface hub **106** configured to facilitate communication between the response computer **104** and the response unit(s) **102**. Even further, the system **100** may include an input interface **108** configured to display a graphical user interface (GUI) **110**, such as a display configured to received touch input, so that the system **100** may receive commands and other types of inputs from a user.

[0027] The system **100** may be activated in various manners. As described in further detail below, the system **100** may be activated autonomously, such as in response to the detection of an active assailant by sensor(s) coupled to one or more of the units **102** or by an external detection unit **114** disposed in an environment (e.g., a school) equipped with the system **100**. The sensor(s) and/or external detection unit **114** may include a camera, a LIDAR device, Additionally or alternatively, the system **100** may include a physical switch that, when actuated, causes the system **100** to activate.

[0028] Although FIG. **1** depicts the elements of the active assailant response system **100**, the figure only includes one "zone" of the system **100**. In some implementations, only one such zone may be provided. In other implementations, however, the system **100** may include multiple zones **100** as the environment in which the system **100** is disposed may be larger or include different sections that are at least somewhat separated from one another. For example, as shown in FIG. **2**, the active assailant response system **100** may be disposed in a school to combat school shootings. In the illustrated example, the school includes Zones **1-7**, and the response units **102** of the system **100** are disposed in each of the zones as shown by the star icons. Alternatively, the system **100** may be configured to protect some of the zones rather than all of the zones. Further, the system **100** may be configured to corral the active assailant towards and into one of the zones, such as Zone **1**. This allows the system **100** to move the active assailant towards zones in which there are no bystanders, and away from zones that do include bystanders. To this end, although not shown, the system **100** may be in communication with door mechanism of doors that separate the zones or parts of one zone, and system **100** may send a signal to the door mechanism to close and/or lock the door separating the first and second zones in response to the active assailant having been corralled into one of the first and second zones of the environment.

[0029] Referring to FIG. **3**, a block diagram illustrating various components of an implementation of the active assailant response system **100** configured to protect multiple zones of a building environment is shown. In this implementation, there are three zones, Zones **1-3**, but it will be appreciated that the ideas discussed in reference to this figure may also be applied to any number of zones.

[0030] Starting with the specific zones, each of Zones **1-3** is shown to include the components of the system **100** shown in FIG. **1**. More specifically, Zone **1** may include a first response unit **102**A (or multiple units **102**A), a first response computer **104**A, a first interface hub **106**A, means for displaying a first GUI **110**A, a first physical switch **112**, and a first external detection unit **114**. Corresponding elements may be disposed in Zone **2** and Zone **3**, such as second and third response units **102**B, **102**C (or multiple units **102**B, **102**C in each zone), second and third response computers **104**B, **104**C, second and third interface hubs **106**B, **106**C, means for displaying second and third GUIs **110**B, **110**C, second and third physical switches **112**B, **112**C, and/or second and third external detection units **114**B, **114**C. Each of the zone-specific components, such as the first response computer **104**A, are labeled with a letter (A-C) to indicate that it is zone-specific. That said, it should be understood that the elements labeled with a letter are similar in structure and configuration to those labeled without a letter. For example, the first response computer **104**A may be like the response computer **104** shown and described in reference to FIG. **1**.

[0031] As for the active assailant response system **100** as a whole, the system **100** may include means for controlling each zone. In the illustrated implementation, the system **100** includes a master computer **120** configured to communicate with the response computers **104**A, **104**B, **104**C of each zone over a private network. Additionally, the system **100** may include a physical switch

122 and/or GUI **124** in direct communication with the master computer **120**, and these elements **122**, **124** may be used to control the master computer **120** (and the components of each zone by extension). Further, at least one of the computers **104**A, **104**B, **104**C, **120** may include a memory device (not shown), and the memory device may include a map of the environment. The map of the environment may be used for various purposes, such as to corral the active assailant as described in more detail below.

[0032] The response units **102** include a turret unit configured to identify, track, and subdue an active assailant, and may further include a traversal unit configured to move the turret unit. If movement of the response unit **102** is not desired or not necessary, the response unit **102** may include only a turret unit fixably coupled to a wall or other surface. Additionally or alternatively, the response unit **102** may include a traversal unit where it is necessary or desired for the response unit **102** to move.

[0033] Referring to FIGS. **4**A-**5**F, two implementation of a turret unit **130** are shown. First, looking to FIGS. **4**A-**4**C, a first implementation of the turret unit **130** is shown in detail. In this implementation, the turret unit **130** includes a nonlethal device **132** for incapacitating/subduing the active assailant, a nonlethal actuator **134** for activating the nonlethal device **132**, a camera **136** for tracking/identifying the active assailant, and a LIDAR unit **138** also for tracking/identifying the active assailant. The nonlethal device **132** may include a weapon that is intended to inflict temporary pain or discomfort unto an individual, such as a pepper spray dispenser, a pepperball gun, a rubber bullet gun, a bean bag gun, a taser, and a flashbang launcher. The LIDAR unit **138** may be a Terabee Distance Sensor Modules LiDAR ToF Range Find (TR-EVO-15M-12C), which has both depth perception and thermal imaging capabilities. The camera **136** and the LIDAR unit **138** are coupled to a first bracket **140**, while the nonlethal device **132** and nonlethal actuator **134** are coupled to a second bracket **142**. Both of the first and second brackets **140**, **142** are coupled to a third bracket **144**. In order to allow the turret unit **130** to aim the nonlethal device **132** and keep track of the active assailant using the camera **136** and/or the LIDAR unit **138**, the turret unit **130** may include a first motor **146** and a second motor **148**. In the illustrated implementation, the first motor **146** is housed within the first bracket **140** and allows the turret **130** to pivot about a pitch axis PA, while the second motor **148** is received by the third bracket **144** and permits the turret **130** to pivot about a yaw axis YA. Finally, the turret unit **130** may include a fourth bracket **150** arranged to house the second motor **148**, and a fifth bracket **152** for coupling the turret unit **130** to a wall, a ceiling, a floor, or a traversal unit. Although not shown in FIGS. **4**A-**4**C, the turret unit **130** may further include a unit computer as described below.

[0034] Second, focusing on FIGS. **5**A-**5**F, a second implementation of the turret unit **130** is shown. Like the prior implementation, the turret **130** of FIGS. **5**A-**5**F includes the nonlethal device **132**, the nonlethal actuator **134**, the camera **136**, and the fifth bracket **152** for coupling the turret unit **130** to a wall, a ceiling, a floor, or a traversal unit. Although not shown, this implementation of the turret unit **130** may also include the LIDAR unit **138** disposed adjacent to the camera **136**. Differently here, the turret unit **130** of FIGS. **5**A-**5**F includes a housing **154** which surrounds most of the remaining components of the turret **130**. This may be advantageous for scenarios where it is desired or necessary to conceal the response unit **102** from view of individuals in the environment in which the system **100** is disposed. For example, where the environment is a school, and it is not desirable for the unit **102** to be visible to children. The housing **154** may be coupled to the fifth bracket **152**, while an arm **153** may extend down from the housing **154** and support the nonlethal device **132**, the nonlethal actuator **134**, and the camera **136**.

[0035] As shown in FIGS. **5**E and **5**F, the housing **153** encloses the motors **146**, **148**, and the motors **146**, **148** are each connected to the arm **153**. Similar to the previous implementation, the first motor **146** allows the turret unit **130** to pivot the nonlethal device **132** and the camera **136** about the pitch axis PA, while the second motor **148** enables rotation about the yaw axis YA. Further, the housing **154** may surround an external input interface **156** and a power source **158**. The external input interface **156** may include at least one USB port, HDMI port, ethernet port, or other input means for receiving data and communications from other devices. The power source **158** may include a battery (not shown), or may be configured to receive a power cord, such as through a port **160** defined by the housing **154**. Even further, to control the various components, the turret unit **130** may include a unit computer **162**, a targeting computer **164**, and a actuator controller **166**. The unit computer **162** may be connected to each of the external input interface **156**, the power source **158**, the targeting computer **164**, and the controlling computer **166**. This way, the unit computer **162** may receive power from the power source **158**, pass power to each of the other computers **164**, **166**, receive data and communications from devices connected to the external input interface **156**, and facilitate communication between each of the elements of the turret unit **130**. While the unit computer **162** may be configured to control the overall operation of the turret unit **130**, the targeting computer **164** may be in communication with the camera **136** (and/or the LIDAR unit **138** depending on the implementation) to provide means of tracking active assailants as further described below. The actuator controller **166**, on the other hand, may be in communication with the motors **146**, **148**, and may cause the motors **146**, **148** to pivot the turret unit **130** about the pitch and yaw axes PA, YA, such as in response to movement of the active assailant captured by the camera **136**. In some implementations, the targeting computer **164** and the actuator controller **166** are implemented as part of the unit computer **162**.

[0036] Referring to FIGS. **6**A-**9**D, different forms of a traversal unit are illustrated. As briefly mentioned above, the response unit(s) **102** may include the turret unit **130** and, optionally, a traversal unit configured to move the turret unit **130** via rail, air, cable, or ground. To this end, a rail traversal unit **170** is shown in FIGS. **6**A-**6**D, an aerial traversal unit **190** is shown in FIGS. **7**A-**7**D, a cable traversal unit **210** is shown in FIGS. **8**A-**8**D, and a wheeled traversal unit **230** is shown in FIGS. **9**A-**9**D. As described above, each zone may include any number of response units **102** equipped with any type of traversal unit (or even coupled to a wall without a traversal unit). It is contemplated for the response unit(s) **102** to include either implementation of the turret unit **130** in combination with any type of traversal unit.

[0037] In FIGS. **6**A-**6**D, an implementation of the response unit **102** equipped with the rail traversal unit **170** is shown. The rail traversal unit **170** is configured to move the response unit **102** (i.e., the turret unit **130**) along a straight or curved rail R. The rail traversal unit **170** may include a first base plate **172** to which the turret unit **130** is coupled, a second base plate **173** coupled to the first base plate **173**, a plurality of guide rollers **174** arranged to guide the response unit **102** along the rail R, a plurality of supports **176** coupled to the second base plate **173** and the guide rollers **174** to fix the position of the rollers **174** relative to the rest of the unit **102**, a drive gear **178** configured to engage the rail R, and a drive motor **180** configured to drive the drive gear **178** in order to move the unit **102** along the rail R. Although not shown, the drive motor **180** may be connected to the actuator controller **166** and/or the unit computer **162** of the response unit **102** such that at least one of the computers **162**, **166** may control movement of the unit **102**.

[0038] In FIGS. **7**A-**7**D, an implementation of the response unit **102** equipped with the aerial traversal unit **190** is shown. The aerial traversal unit **190** is configured to move the response unit **102** (i.e., the turret unit **130**) through the air. The aerial traversal unit **190** includes a housing **191**, a frame **192** coupled to the housing **191** and configured to fix the turret unit **130** to the housing **191**, and a plurality of propellers **194** rotatably connected to the housing **191** and configured to generate lift. Although not shown, a power supply and a series of electric motors connected between the power supply and the propellers **194** may be included within the housing **191** for driving the propellers **194** and generating lift. In the illustrated implementation, the aerial traversal unit **190** further includes a secondary sensor package **196** for optically sensing the environment in ways similar to those described in reference to the camera **136** and the LIDAR unit **138** herein.

[0039] In FIGS. **8**A-**8**D, an implementation of the response unit **102** equipped with the cable traversal unit **210** is shown. Similar to the aerial traversal unit **190**, the cable traversal unit **210** is configured to move the response unit **102** (i.e., the turret unit **130**) through the air. Further similar to the aerial traversal unit **190**, the cable traversal unit **210** may include a housing **211**, a frame **212** coupled to the housing **211** and configured to fix the turret unit **130** to the housing **211**, and the secondary sensor package **196**. Differently here, however, the cable traversal unit **210** includes a plurality of receiver rings **214** coupled to the housing **211**. In order to move the cable traversal unit **210** through the air, the unit **210** further includes a plurality of base stations **216** that are meant to be attached to various walls (and/or the ceiling) of the environment in which the system **100** is disposed. Additionally, a plurality of cables **218** extending between each base station **216** and a corresponding receiver ring **214** are controlled by the base stations **216** to pull the cable traversal unit **210** through the air. In some implementations, a plurality of pulleys **220** may be coupled to the wall(s)/ceiling to guide at least a portion of the cables **218**. Finally, although not shown, the cable traversal unit **210** may include electric motors disposed within the base stations **216** and configured to draw the cables **218** towards the base stations **216** to move the unit **210**.

[0040] In FIGS. **9**A-**9**D, an implementation of the response unit **102** equipped with the wheeled traversal unit **230** is shown. The wheeled traversal unit **230** is configured to move the response unit **102** (i.e., the turret unit **130**) across the ground. The wheeled traversal unit **230** may include a housing **231** and a frame **232** coupled to the housing **231** and configured to fix the turret unit **130** to the housing **231**. In order to move the unit **230**, a plurality of wheels **234** rotatably connected to the housing **231** may be provided, each of the plurality of wheels **234** connected to one of a plurality of electric motors **236** disposed within the housing **231**.

[0041] Regardless of implementation or traversal means provided, the response unit **102** communicates with, and is controlled by, the response computer **104** and/or the master computer **120**. Further, the turret unit **130** of the response unit **102** is generally configured to communicate with the traversal unit **170**, **190**, **210**, **230** coupled thereto. This way, the turret unit **130** may receive information from one or both of the computers **104**, **120** and control the traversal unit **170**, **190**, **210**, **230** in response to the received information and in order to move the response unit **102**. This is described in more detail in reference to FIG. **10** below.

[0042] Referring to FIG. **10**, a block diagram showing connections between components of the response unit **102** according to some implementations is shown. As shown, the response unit **102** includes the turret unit **130** and, optionally, one implementation of the traversal unit **170**, **190**, **210**, **230**. The turret unit **130** includes the unit computer **162**, which may receive power from the power source **158**, communicate with devices connected to the external input interface **156**, and communicate with the targeting computer **164** to identify/track active assailants. More specifically, in response to signals received by the targeting computer **164**, the unit computer **162** may then communicate with the actuator controller **166** to aim the unit **102** via the motors **146**, **148**, activate the nonlethal device **132** with the nonlethal actuator **134**, and/or move the unit **102** by causing the movement actuators **180**, **195**, **216**, **236** to drive the movement devices **178**, **194**, **218**, **234**. In some implementations of the response unit **102**, the unit is mounted to a wall/ceiling and does not include the traversal unit **170**, **190**, **210**, **230**.

[0043] As briefly described above, the system **100** may be controlled by at least one of the physical switch **112** and/or input interface **108** configured to display the GUI **110**. To this end, the physical switch **112** and the GUI **110** are shown in more detail in FIGS. **11** and **12**.

[0044] First, looking to FIG. **11**, the physical switch **112** may include a lever **113** or other suitable means of receiving an activation input from a user. Further, in some implementations, the physical switch **112** may include a scanning device **115** configured to authenticate the user prior to allowing activation of the system **100**. The scanning device **115** may be configured to authenticate the user by scanning a code presented by the user, such as a barcode or QR code. In other implementations, the scanning device **115** may be an RFID reader configured to communicate with an RFID tag carried by the user. Second, referring to FIG. **12**, the GUI **110** may display a plurality of buttons **111** that may be pressed by the user to control the system **100**. Like the example described above, the input interface **108** may be a touch-enabled display, and the user may interact with the buttons **111** by engaging the touch-enabled display. The buttons **111** may activate/deactivate the system **100** as a whole or activate/deactivate specific zones of the system **100**.

II. Method Overview

[0045] The active assailant response system **100** be controlled according to a number of methods as described herein. These methods are described in this section and in reference to FIGS. **13-16**. Put simply, the methods may be carried out (e.g., by the system **100**) in order to: (1) activate and control a single-zone implementation of the system **100** to corral an active assailant as shown in FIG. **13**, (2) activate and control a multi-zone implementation of the system **100** to corral an active assailant as shown in FIG. **14**, (3) identify and track an active assailant in order to deploy a nonlethal device towards the assailant as shown in FIG. **15**, and (4) update the system **100** as shown in FIG. **16**.

[0046] Referring to FIG. **13**, a flowchart describing a method **300** of controlling a single-zone implementation of the active assailant response system **100** is shown. At **304**, the system is activated. As shown, the system **100** may be activated manually at step **304**A or autonomously at step **304**B. If the system **100** is activated autonomously, such as in response to detection of an active assailant by one of the response units **102** and/or external detection unit **114**, the method **300** may include determining if a user confirms the threat at **306**. If step **306** is included and the user does not confirm the threat, the method **300** may jump to step **328** and the system **100** may shut down. Alternatively, the method may continue to step **308** if the system **100** was activated manually at step **304**A, the user confirms the threat at step **306**, or if the system **100** was activated autonomously at step **304**B and step **306** is not included.

[0047] At **308**, the response units **102** are activated and begin searching for the active assailant. For example, if the external detection unit **114** triggered activation of the system **100**, the units **102** may attempt to move towards the external detection unit **114**. As another example, if the system **100** was activated manually from the physical switch **112**, the units **102** may attempt to move towards the switch **112**. As the response units **102** move towards the potential location of the threat and/or once at least one of the units **102** to the potential location, the units **102** attempt to identify the threat at **312**. This may include using one or both of the camera **136** and the LIDAR unit **138** to detect a weapon, such as according to the method described in reference to FIG. **15** below.

[0048] At **316**, the system **100** determines if an active assailant was detected at **312**. If not, the method **300** proceeds to step **328** and the system **100** shuts down. If the active assailant is detected, however, the method moves to step **320**. Then, at **320**, at least one of the units **102** deploy the nonlethal device **132** toward the active assailant. In some implementations, the system **100** is configured to use the units **102** to corral the assailant towards a predetermined location, such as outside the building in which the system **100** is disposed. As such, the method **300** may include determining if the assailant has been corralled at step **324**. If the assailant has not been corralled, the method **300** returns to step **320**, at which point the system **100** controls the units **102** to continue deploying the nonlethal device **132** toward the assailant to urge them toward the predetermined location. Once the assailant is determined to have been corralled at step **324**, the method **300** moves to step **328** and the system **100** is shut down.

[0049] Referring to FIG. **14**, a flowchart describing a method **400** of controlling a multi-zone implementation of the active assailant response system **100** is shown. Similar to the single-zone method **300**, the multi-zone method **400** begins with system activation at step **404**. If the system **100** is activated autonomously, such as in response to detection an active assailant by one of the response units **102** and/or external detection unit **114**, the method **400** may include determining if a user confirms the threat at **406**. If the threat is not confirmed at **406**, the method **400** may move to step **432** and the system **100** may shut down. Instead, if the system **100** was activated manually at **404**A, if the user confirms the threat at **406**, or if the method **400** does not include step **406**, the method **400** moves to step **408**.

[0050] At **408**, a first zone of the system **100**, such as Zone **1**, is activated by the corresponding response computer **104**. The first zone corresponds to the zone in which the physical switch **112**/GUI **110** that was used to manually activate the system **100** at **404**A is located or in which the activate assailant was detected/confirmed at **404**B and/or at **406**. Assuming that Zone **1** is the first zone, the first response computer **104**A transmits an activation signal to the master computer **120** at **412**. In response, the master computer **120** transmits an activation signal to the response computers **104**B, **104**C associated with the other zones, such as Zone **2** and Zone **3**.
[0051] Once activated, the response computers **104**A, **104**B, **104**C activate their respective response units **102**A, **102**B, **102**C at **420**. The response units **102**A, **102**B, **102**C then begin searching for the active assailant at **424**, such as in ways similar to those described in reference to step **312** above. The method **400** continues to step **426**, at which point the system **100** determines if the assailant can be detected. If not, the method **400** jumps to step **432** and the system **100** is shut down. If the active assailant is detected, however, the method moves to step **428**. At **428**, nonlethals **132** are deployed to corral the active assailant, such as towards a predetermined zone or out of the environment. Then, at **430**, the system **100** determines if the assailant has been corralled into the predetermined zone or out of the environment. If not, the method returns to step **428**. If the active assailant has been corralled, the method **400** moves to step **432** and the system **100** shuts down.
[0052] Further, in both of the methods **300**, **400** described above, the response units **102** are tasked with locating/tracking the active assailant in order to deploy the nonlethal device **132** to corral/subdue the assailant. This occurs, for example, during steps **308** through **320** of the single-zone method **300** and/or steps **416** through **430** of the multi-zone method **400**. To improve the capabilities of the response units **102** in locating/tracking the assailant, a machine learning algorithm may be used. The machine learning algorithm may assist the unit **102** in differentiating active assailants from potential victims, as well as determining when a person becomes an active assailant (e.g., by drawing a weapon).
[0053] Referring to FIG. **15**, a flowchart describing a method **500** of controlling one of the response units using a targeting algorithm **510** and an actuator control algorithm **550** is shown. The method **500** is described as being carried out by one response unit **102**, but the method **500** may be carried out by each response unit **102** of the system **100**. Further, the method **500** is described as being carried by the unit computer **162**, the targeting computer **164**, and the actuator controller **166**, but the method **500** may also be executed by the unit computer **162** alone.
[0054] Starting with the targeting algorithm **510**, at **512**, optical data is received by the targeting computer **164**. The optical data may be generated by at least one of the camera **136** and the LIDAR unit **138**. Subsequently, at **516**, the targeting computer **164** may apply a weapon detection program to the optical data in order to detect weapons present in the optical data. The weapon detection program may be a machine learning algorithm, such as one trained on image/video data that depicts weapons. In one implementation, the weapon detection program is a trained version of the YOLOv8 algorithm released in 2023 by Ultralytics. The targeting computer **164** may require the weapon to be detected in at least three frames of optical data over a predetermined time period, such as one second, prior to confirming that a weapon has been detected at **516**. Once a weapon is detected at **516**, the method **500** moves to **520**. At **520**, a camera motion estimator is applied to the optical data to estimate a direction of movement of the weapon. With the weapon detected and its motion estimated, an object tracking model may be applied at **524** in order to keep track of the position of the weapon. This may include generating a bounding box around the weapon and/or associated active assailant, and overlaying the bounding box on the optical data so as to surround the weapon/assailant as they appear in the optical data.
[0055] Once the weapon is being tracked in the optical data, a weapon prediction model may be applied at **528**. The weapon prediction model is configured to determine an expected relationship between the weapon and at least a portion of the body of the active assailant (e.g., the head of the assailant). In one case, this may include determining a type of the weapon, such as a pistol or a

rifle, and determining an expected spatial relationship between a weapon of this type and the head of the assailant. In another case, this may include determining a pose of the weapon, such as an aiming pose or a cross-body pose, and determining an expected position of head of the assailant based on the pose of the weapon. Based on an output from the weapon prediction model, the location of the active assailant may be determined at **532**. Then, at **536**, a unit position and an aim angle of the response unit **102** may be calculated/determined at **536**. The unit position and aim angle may correspond to those at which the unit **102** may deploy the nonlethal device **132** towards the active assailant, such as at the head of the active assailant. Further, the unit position may be determined according to how the active assailant should be corralled. For example, if the active assailant is in Zone **1**, the unit position may be determined according to whether the active assailant should be corralled towards or away from Zone **1**. The unit position and aim angle may be output to the actuator controller **166** at **540**. The method **500** then continues to the actuator control algorithm **550**.

[0056] Starting with step **552**, the actuator control algorithm **550** begins by controlling the movement actuator **180**, **195**, **216**, **236** to cause the movement device **178**, **194**, **218**, **234** to move the response unit **102** to the unit position output by the targeting algorithm **510** at **540**. At **556**, the actuator controller **166** controls the aiming actuators **146**, **148** to match the aim angle output by the targeting algorithm **510** at **540**. The response unit **102** may determine if the target (e.g., the active assailant) has moved since step **540**. If the target has not moved, the actuator controller **166** may cause the nonlethal actuator **134** to activate the nonlethal device **132** towards the active assailant. That said, if the target has moved since step **540**, the actuator controller **166** may minimize any disparity between a current aim angle of the unit **102** and an updated aim angle of the unit **102** at **560**. For example, in some implementations, the targeting algorithm **510** is continuously executed and periodically sends the updated aim angle to the actuator controller **166** during the method **500**. Once the disparity is minimized, such as by changing the current aim angle to match the updated aim angle, the nonlethal device **132** is deployed at **564**.

[0057] In some implementations, the active assailant response system **100** may be configured to be periodically updated. This may be advantageous where improvements to the various methods and algorithms **300**, **400**, **500**, **510**, **550** are desired after the system **100** has been installed in a building or other type of environment. This may be accomplished by sending the updates to the master computer **120** (and/or the response computer(s) **104**. In addition to updating the operation of the master computer **120** and/or response computer **104**, the individual response units **102** may also be updated.

[0058] In any of the methods **300**, **400**, **500** described above, the response units **102** may be controlled to corral the active assailant into/out of one of the zones and/or out of the environment. To this end and as briefly described above, the active assailant response system **100** may include a map of the environment (e.g., stored on at least one of the computers **104**, **120**). The map of the environment may include bounds of the environment, boundaries defining the various zones of the environment, locations of doors and walls, locations of natural barricades, positions of the units **102** within the environment, and/or other details about the environment. The system **100**, such as the response computer(s) **104**, **104**A, **104**B, **104**C and/or the master computer **120**, may thus determine how to instruct the response units **102** based on the map of the environment. Additionally, the system **100** may determine a corral position which corresponds to the position where the assailant should be corralled to. The corral position may include an entire zone, a specific position within the zone, or an area within the zone. The unit position may be determined based on the corral position and the map of the environment. Further, the system **100** may may be in communication with door mechanism of doors that separate the zones or parts of one zone, and system **100** may send a signal to the door mechanism to close and/or lock the door separating the first and second zones in response to the active assailant having been corralled into the corral position.

[0059] In one example, the system **100** may determine that the active assailant should be corralled towards or away from the first zone, and determine the unit position based on whether the active assailant should be corralled towards or away from the first zone. In this example, the system **100** may determine that the assailant should be corralled towards the first zone based on the map of the environment, as well as determine the unit position based on the map of the environment and the determination that the assailant should be corralled towards the first zone. In another example where the environment is smaller and includes a single zone, such as a convenience store, the system **100** determine that the assailant should be corralled against a corral position corresponding to a specific wall within the environment based on the map, and instruct the units **102** to move to unit positions that position the active assailant between the units **102** and the specific wall.
[0060] Referring to FIG. **16**, a flowchart describing a method **600** of updating at least one of the response units **102** is shown. Starting at **604**, the update may be provided to the master computer **120** in a multi-zone implementation of the system **100** or to the response computer **104** in a single-zone implementation of the system **100**. If the system **100** is a single-zone implementation, the method **600** jumps to step **612**. That said, if the system **100** is a multi-zone implementation, such as the implementation depicted in FIG. **3**, the method **600** may continue to step **608** and distribute the update to the response computers **104**. Then, at **612**, an update procedure may be sent to the response units **102** (e.g., by one of the response and master computers **104**, **120**). The update procedure includes instructions that, when executed by the unit computer **162**, prepare the units **102** to receive and apply the update. Once the units **102** have been prepared for the update, the update is applied to the units **102** at step **616**. The units **102** may then restart at step **620**, as well as send an update confirmation to the master computer **120** (e.g., via the corresponding response computer **104**).
III. Exemplary System Layouts
[0061] Many arrangements of the various components of the active assailant system **100** are contemplated. Since every such arrangement cannot be described herein, two exemplary arrangements of the system **100** are depicted in FIGS. **17** and **18**.
[0062] Referring to FIG. **17**, a first arrangement of response unit disposed in various zones of a small school is shown. As shown, response units **102** equipped with the rail traversal unit **170** are installed in hallways, and response units **102** including only the turret unit **130** are installed near each door/entryway.
[0063] Referring to FIG. **18**, a second arrangement of response unit disposed in various zones of a large school is shown. Similar to the arrangement of the system **100** shown in FIG. **17**, response units **102** equipped with the rail traversal unit **170** are install in the hallways, and turret unit **130** are installed near each door/entryway. Additionally here, response units **102** equipped with cable traversal units **210** are installed in large spaces that may include ground obstacles, such as gymnasiums and lunchrooms. For large open spaces that do not include ground obstacles, response units **102** equipped with the wheeled traversal unit **230** are installed therein. Finally, response units **102** having the aerial traversal unit **190** are disposed outside of the school.
[0064] Several embodiments have been discussed in the foregoing description. However, the implementations discussed herein are not intended to be exhaustive or limit the filter assembly to any particular form factor. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the system may be practiced otherwise than as specifically described.

Claims

1. An active assailant response system comprising: a set of response units configured to deploy a nonlethal device, the set of response units including: a first response unit arranged in a first zone of an environment, and a second response unit arranged in a second zone of the environment; a set of detection units configured to identify one or more active assailants within the environment, the set of detection units including: a first detection unit arranged in the first zone of the environment, and a second detection unit arranged in the second zone of the environment; a computer in communication with the set of response units and the set of detection units, the computer being configured to: determine that at least one active assailant is present within one of the first and second zones based on signals received from the set of detection units, and activate the set of response units in response to the at least one active assailant being present within one of the first and second zones.

2. The active assailant response system of claim 1, wherein the first response unit includes the first detection unit and the second response unit includes the second detection unit.

3. The active assailant response system of claim 2, wherein the set of detection units further includes a third detection unit coupled to a wall of the environment, the wall being in one of the first and second zones of the environment.

4. The active assailant response system of claim 2, wherein at least one of the first and second response units includes a turret unit, the turret unit comprising: a housing; one of the first and second detection units, the one of the first and second detection units including at least one of a camera and a LIDAR unit coupled to the housing; a nonlethal actuator coupled to the housing; and a nonlethal device coupled by the housing and arranged in contact with the nonlethal actuator.

5. The active assailant response system of claim 4, wherein the nonlethal device includes at least one of a pepper spray dispenser, a pepperball gun, a rubber bullet gun, a bean bag gun, a taser, and a flashbang launcher.

6. The active assailant response system of claim 4, wherein the at least one of the first and second response units further includes a unit computer coupled to the housing, the unit computer configured to: receive optical data from the at least one of the camera and the LIDAR unit; identify an active assailant based on the optical data; and activate the nonlethal actuator in response to the active assailant being identified.

7. The active assailant response system of claim 6, wherein the at least one of the first and second response units further includes at least one aiming actuator configure to pivot the response unit in at least one of a first and second degree of freedom, and the unit computer is configured to cause the at least one aiming actuator to keep the nonlethal device aimed at the active assailant.

8. The active assailant response system of claim 7, wherein the at least one of the first and second response units further includes a traversal unit configured to move the at least one of the first and second response units through the environment.

9. The active assailant response system of claim 8, wherein the traversal unit includes one of a rail traversal unit, an aerial traversal unit, a cable traversal unit, and a wheeled traversal unit.

10. The active assailant response system of claim 8, wherein the traversal unit is in communication with the unit computer, and the unit computer is further configured to: determine a location of the active assailant; determine a unit position corresponding to a position of the at least one of the first and second response units relative to the environment; determine an aim angle corresponding to an angle between the at least one of the first and second response units and the active assailant at which the nonlethal device should be deployed; cause the traversal unit to move the at least one of the first and second response units to the unit position; cause the at least one aiming actuator to match a current angle of the nonlethal device to the aim angle; and cause the nonlethal actuator to activate the nonlethal device in response to the at least one of the first and second response units being at the unit position and aiming towards the aim angle.

11. The active assailant response system of claim 10, where the unit computer is further configured to: determine that the active assailant is present within the first zone; determine that the active assailant should be corralled towards or away from the first zone; and determine the unit position based on whether the active assailant should be corralled towards or away from the first zone.

12. The active assailant response system of claim 6, wherein the unit computer is configured to apply a machine learning algorithm to the optical data to identify the active assailant.
13. The active assailant response system of claim 1, further comprising at least one of a physical switch and a graphical user interface in communication with the computer and configured to activate the set of response units in response to input from a user.
14. The active assailant response system of claim 13, wherein the physical switch includes a scanning device configured to authenticate the user, and the computer is configured to deactivate the set of response units if the user is not authenticated.
15. The active assailant response system of claim 14, wherein the scanning device includes one of an RFID reader, a barcode scanner, and a QR code reader.
16. The active assailant response system of claim 1, wherein the computer is configured to: activate the first response unit in response to the active assailant being present within the first zone of the environment; and activate the second response unit in response to the active assailant being present within the second zone of the environment.
17. The active assailant response system of claim 1, wherein the computer is configured to: determine that the active assailant should be corralled towards the first zone of the environment; and instruct the set of response units to corral the active assailant towards the first zone of the environment.
18. The active assailant response system of claim 1, further comprising: an external detection unit in communication with the computer and arranged in at least one of the first and second zones of the environment; and an input interface in communication with the computer and arranged in the at least one of the first and second zones of the environment; wherein the computer is configured to: receive a signal from the input interface, determine if the active assailant has been confirmed by a user based on the signal, and deactivate the set of response units in response to the active assailant not being confirmed.
19. The active assailant response system of claim 1, further comprising: a memory device in communication with the computer and having a map of the environment stored thereon, the map including boundaries defining the first and second zones of the environment; wherein the computer is configured to cause the set of response units to corral the active assailant towards one of the first and second zones of the environment based on the map.
20. The active assailant response system of claim 19, wherein the computer is in communication with a door mechanism of a door separating the first and second zones, and the computer is configured to send a signal to the door mechanism to close and/or lock the door separating the first and second zones in response to the active assailant having been corralled into one of the first and second zones of the environment.
21. An active assailant response system comprising: a response unit arranged in an environment, the response unit including: a traversal unit configured to move the response unit about the environment, a turret unit coupled to the traversal unit and configured to pivot about at least two degrees of freedom, a nonlethal device supported by the turret unit, and a detection unit supported by the turret unit configured to identify one or more active assailants within the environment; a computer in communication with the response unit, the computer being configured to: determine that at least one active assailant is present within the environment based on signals generated by the detection unit, and activate the response unit in response to the at least one active assailant being present within the environment.
22. The active assailant response system of claim 21, wherein: the system further comprises a memory unit in communication with the computer and having a map of the environment stored thereon; and the computer is further configured to: determine a corral position to which the active assailant should be corralled, determine a unit position based on the corral position and the map of the environment, and cause the response unit **102** to move to the unit position in order to corral the active assailant.

23. The active assailant response system of claim 22, wherein the computer is configured to: determine a position of the active assailant, and determine the unit position such that the position of the active assailant is between the unit position and the corral position.
24. The active assailant response system of claim 21, wherein the computer is in communication with a door mechanism of a door within the environment, and the computer is configured to send a signal to the door mechanism to close and/or lock the door in response to the active assailant having been corralled into the corral position.
25. The active assailant response system of claim 21, further comprising an external detection device coupled to a wall of the environment.
26. The active assailant response system of claim 21, wherein the nonlethal device includes at least one of a pepper spray dispenser, a pepperball gun, a rubber bullet gun, a bean bag gun, a taser, and a flashbang launcher.
27. The active assailant response system of claim 21, wherein the turret unit comprises: a housing; at least one aiming actuator configured to pivot the turret unit about the at least two degrees of freedom; and a nonlethal actuator coupled to the housing and configured to activate the nonlethal device.
28. The active assailant response system of claim 27, wherein the at least one of the first and second response units further includes a unit computer coupled to the housing, the unit computer configured to: receive optical data from the detection unit; identify an active assailant based on the optical data; and activate the nonlethal actuator in response to the active assailant being identified.
29. The active assailant response system of claim 28, wherein the unit computer is configured to cause the at least one aiming actuator to keep the nonlethal device aimed at the active assailant.
30. The active assailant response system of claim 21, wherein the traversal unit includes one of a rail traversal unit, an aerial traversal unit, a cable traversal unit, and a wheeled traversal unit.
31. The active assailant response system of claim 30, wherein the traversal unit is in communication with the unit computer, and the unit computer is further configured to: determine a location of the active assailant; determine a unit position corresponding to a position of the at least one of the first and second response units relative to the environment; determine an aim angle corresponding to an angle between the at least one of the first and second response units and the active assailant at which the nonlethal device should be deployed; cause the traversal unit to move the at least one of the first and second response units to the unit position; cause the at least one aiming actuator to match a current angle of the nonlethal device to the aim angle; and cause the nonlethal actuator to activate the nonlethal device in response to the at least one of the first and second response units being at the unit position and aiming towards the aim angle.
32. The active assailant response system of claim 31, wherein the unit computer is configured to apply a machine learning algorithm to the optical data to identify the active assailant.
33. The active assailant response system of claim 21, further comprising at least one of a physical switch and a graphical user interface in communication with the computer and configured to activate the response unit in response to input from a user.
34. The active assailant response system of claim 33, wherein the physical switch includes a scanning device configured to authenticate the user, and the computer is configured to deactivate the response unit if the user is not authenticated.
35. The active assailant response system of claim 34, wherein the scanning device includes one of an RFID reader, a barcode scanner, and a QR code reader.
36. The active assailant response system of claim 21, further comprising: an external detection unit in communication with the computer and arranged in the environment; and an input interface in communication with the computer and arranged in the at least one of the first and second zones of the environment; wherein the computer is configured to: receive a signal from the input interface, determine if the active assailant has been confirmed by a user based on the signal, and deactivate the set of response units in response to the active assailant not being confirmed.